Flutter Viva Questions – All 13 Practicals

Practical 1: Installation of Flutter

1. What is Flutter?

Flutter is an open-source UI framework developed by Google. It is used to create mobile, web and desktop applications.

2. What is Dart?

Dart is the programming language used to develop Flutter applications.

3. Who developed Flutter?

Google developed Flutter.

4. What is the use of Flutter Doctor?

flutter doctor checks whether Flutter and its required tools are installed correctly.

5. What is an SDK?

SDK stands for Software Development Kit. It contains tools required to develop applications.

6. What is an emulator?

An emulator is software that behaves like a mobile device and helps us test applications.

7. What is the use of VS Code?

VS Code is a code editor used to write and run Flutter programs.

Practical 2: Simple Flutter Demo Page

8. What is a widget?

A widget is the basic building block of a Flutter application.

9. What is main()?

It is the starting point of a Dart program.

10. What is runApp()?

It starts the Flutter application and displays the given widget.

11. What is MaterialApp?

MaterialApp provides the basic structure and Material Design features for a Flutter application.

12. What is Scaffold?

Scaffold provides the basic page layout, such as AppBar, Body and FloatingActionButton.

13. What is AppBar?

AppBar is the top section of an application that displays a title or actions.

14. What is the use of Text widget?

It displays text on the screen.

15. What is Center widget?

It places its child widget at the center.

Practical 3: Custom Widgets

16. What is a Container?

Container is a widget used to create a box with properties like height, width, color and padding.

17. What is an Icon widget?

It displays graphical symbols such as stars, hearts and home icons.

18. What is an ElevatedButton?

It creates a button with an elevated appearance.

19. What is the difference between Text and Container?

Text displays words, whereas Container creates a box that can hold other widgets.

20. What is a child widget?

A child is a widget placed inside another widget.

Practical 4: Row, Column and Stack

21. What is Row?

Row arranges widgets horizontally.

22. What is Column?

Column arranges widgets vertically.

23. What is Stack?

Stack places widgets on top of one another.

24. What is the difference between Row and Column?

Row arranges widgets horizontally, whereas Column arranges widgets vertically.

25. What is the use of mainAxisAlignment?

It controls the alignment of children along the main axis.

26. What is crossAxisAlignment?

It controls the alignment of children along the cross axis.

27. What is the main axis of Row?

The horizontal axis.

28. What is the main axis of Column?

The vertical axis.

Practical 5: Responsive UI

29. What is Responsive UI?

Responsive UI is a design that adjusts according to different screen sizes.

30. What is MediaQuery?

MediaQuery provides information about the screen size and other device properties.

31. What is the use of size.width?

It returns the screen width.

32. What is the use of size.height?

It returns the screen height.

33. Why do we use responsive design?

To make applications look properly on different devices.

Practical 6: Dart Programming Basics

34. What is a variable?

A variable is a named memory location used to store data.

35. What is int?

int is a data type used to store whole numbers.

36. What is String?

String is a data type used to store text.

37. What is double?

double is used to store decimal numbers.

38. What is bool?

bool stores either true or false.

39. What is an if statement?

An if statement executes code when a condition is true.

40. What is a loop?

A loop is used to execute a block of code repeatedly.

41. What is a for loop?

A for loop repeats code based on a specified condition.

42. What is print()?

print() displays output in the console.

43. What is the difference between var and dynamic?

var determines the variable's type from its initial value. dynamic allows the variable to hold values of different types.

44. Is const mandatory in Flutter?

No. const is optional when a constant expression is being used, but it is useful for fixed values and widgets.

Practical 7: Counter App using StatefulWidget

45. What is StatefulWidget?

StatefulWidget is a widget whose state can change during the application's lifetime.

46. What is StatelessWidget?

StatelessWidget is a widget that does not manage changing state.

47. What is setState()?

setState() tells Flutter that the state has changed and the UI needs to be rebuilt.

48. Why do we use setState() in the counter app?

We use it to update the counter value on the screen.

49. What is the initial value of the counter?

Zero.

50. What does count++ mean?

It increases the count value by one.

51. What happens when we click the Add button?

The counter increases by one and the screen displays the updated value.

52. What is the difference between StatefulWidget and StatelessWidget?

StatefulWidget manages changing state, whereas StatelessWidget does not manage changing state.

Practical 8: Student Card using StatelessWidget

53. What is a Card widget?

Card is a Material Design widget used to display related information inside a box.

54. What is Padding?

Padding adds empty space around a widget's content.

55. What is the use of EdgeInsets.all(20)?

It adds 20 logical pixels of padding on all four sides.

56. Why do we use StatelessWidget for the student card?

Because the student details displayed in this program do not change.

57. Can StatelessWidget display dynamic information?

Yes. It can display information passed to it, but it does not manage changing state by itself.

Practical 9: Custom Button

58. What is onPressed?

onPressed defines the action performed when a button is clicked.

59. What is styleFrom()?

It is used to customize the appearance of buttons.

60. How do you change the button background color?

By using the backgroundColor property.

61. What is a callback function?

A callback is a function passed to another function or widget to be executed later.

62. What happens when onPressed is null?

The button becomes disabled.

Practical 10: Navigation Between Screens

63. What is navigation in Flutter?

Navigation is the process of moving from one screen to another.

64. What is Navigator?

Navigator manages the stack of screens in a Flutter application.

65. What is Navigator.push()?

It opens a new screen and adds it to the navigation stack.

66. What is Navigator.pop()?

It closes the current screen and returns to the previous screen.

67. What is MaterialPageRoute?

It creates a route to navigate to another screen using Material Design transitions.

68. What is context?

BuildContext represents the location of a widget in the widget tree.

Practical 11: Form Validation

69. What is Form widget?

Form is a widget used to group and manage form fields.

70. What is TextFormField?

TextFormField is used to accept user input and supports validation.

71. What is validation?

Validation checks whether the entered data is correct.

72. What is GlobalKey?

GlobalKey uniquely identifies a widget and can provide access to its state.

73. What is validator?

Validator checks the input and returns an error message if the input is invalid.

74. What does isEmpty mean?

It checks whether a string contains no characters.

75. What does validate() do?

It checks all form fields using their validator functions.

76. What does returning null in validator mean?

It means the entered value is valid.

Practical 12: Implicit Animation

77. What is animation?

Animation is the process of creating movement or visual changes on the screen.

78. What is implicit animation?

Implicit animation automatically animates changes in widget properties.

79. What is AnimatedContainer?

AnimatedContainer is a widget that automatically animates changes in its properties.

80. What is Duration?

Duration specifies how long an animation takes.

81. What is GestureDetector?

GestureDetector detects user actions such as tapping, dragging and swiping.

82. What is the use of onTap?

It defines an action when the user taps a widget.

83. What is the difference between Container and AnimatedContainer?

Container changes its properties immediately, whereas AnimatedContainer smoothly animates property changes.

Practical 13: Explicit Animation

84. What is explicit animation?

Explicit animation gives the developer direct control over the animation.

85. What is AnimationController?

AnimationController controls the animation, including its duration and direction.

86. What is Tween?

Tween defines the starting and ending values of an animation.

87. What is FadeTransition?

FadeTransition gradually changes the opacity of a widget.

88. What is opacity?

Opacity determines how transparent or visible a widget is. A value of 0 means invisible, and 1 means fully visible.

89. What is initState()?

initState() is called once when a StatefulWidget's state object is initialized.

90. What is dispose()?

dispose() releases resources when a widget is removed permanently.

91. What is vsync?

Vsync synchronizes animation updates with the screen refresh rate.

92. What is SingleTickerProviderStateMixin?

It provides a ticker for controlling one animation controller.

93. What is controller.forward()?

It starts the animation in the forward direction.

Important common Flutter Viva Questions

These questions can be asked from any practical.

94. Is Flutter a framework or a programming language?

Flutter is a UI framework. Dart is its programming language.

95. What is hot reload?

Hot reload updates code changes in a running application without restarting the entire app.

96. What is hot restart?

Hot restart restarts the application and resets its current state.

97. What is the difference between hot reload and hot restart?

Hot reload preserves the current state in most cases, whereas hot restart resets the state.

98. What is a widget tree?

A widget tree is the hierarchical structure of widgets in a Flutter application.

99. What is the parent widget?

A parent widget is a widget that contains another widget.

100. What is the child widget?

A child widget is a widget placed inside another widget.

101. What is pubspec.yaml?

It is a configuration file that contains project information, dependencies and assets.

102. What is the use of Material Design?

Material Design provides a standard design system for creating attractive user interfaces.

103. What is the difference between hot reload and normal execution?

Hot reload applies code changes quickly to a running app, while normal execution starts the app.

104. What is the default programming language of Flutter?

Dart.

105. Can Flutter be used for Android and iOS?

Yes. Flutter supports both Android and iOS development.